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NOT MEASUREMENT SENSITIVE

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MIL-A-46146B  
AMENDMENT 3  
28 October 1992  
SUPERSEDING  
AMENDMENT 2  
21 January 1992

# MILITARY SPECIFICATION

## ADHESIVES-SEALANTS, SILICONE, RTV, NONCORROSIVE (FOR USE WITH SENSITIVE METALS AND EQUIPMENT)

This amendment forms a part of MIL-A-46146B, dated 21 November 1990 and is approved for use by all Departments and Agencies of the Department of Defense

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### 3.3.2.1.1: Delete and substitute:

"3.3.2.1.1 Heat resistance. The cured silicone adhesive-sealant for Group I and Group II after exposure for  $168 \pm 4$  hours (7 days  $\pm 4$  hours) at  $392 \pm 4^{\circ}\text{F}$  ( $200 \pm 2^{\circ}\text{C}$ ) shall meet the requirements for hardness, tensile strength and elongation as specified in table II when tested as specified in 4.5.3.1.2, except that the tensile strength and elongation for Group II material shall be 250 psi (min) and 300 percent (min), respectively. Group III material after exposure for  $72 \pm 4$  hrs (3 days  $\pm 4$  hrs) at  $600 \pm 4^{\circ}\text{F}$  ( $316 \pm 2^{\circ}\text{C}$ ) shall have a hardness, Shore A (Durometer) of 60 max, tensile strength of 200 psi, min (1379KPa) and elongation of 100% min."

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"TABLE VI. Quality conformance inspection.

| Test<br>Characteristic        | Requirements |                       |             | Test<br>Method         |
|-------------------------------|--------------|-----------------------|-------------|------------------------|
|                               | Group I      | Group II              | Group III   |                        |
| <u>Uncured compound</u>       |              |                       |             |                        |
| Total solids content          | 3.3.1        | 3.3.1                 | 3.3.1       | 4.5.3.2.1              |
| Application rate              | 3.3.1        | ---                   | 3.3.1       | 4.5.3.2.2              |
| Flow                          | 3.3.1        | ---                   | 3.3.1       | 4.5.3.2.3              |
| Viscosity                     | ---          | 3.3.1                 | ---         | ASTM D1084<br>Method B |
| Tack free time                | 3.3.1        | 3.3.1                 | 3.3.1       | 4.5.3.2.5              |
| Corrosion <u>1/</u> <u>2/</u> | 3.3.1.1 (a)  | 3.3.1.1 (a)           | 3.3.1.1 (a) | 4.5.3.2.6.1            |
| <u>Cured compound</u>         |              |                       |             |                        |
|                               |              | <u>Groups I - III</u> |             |                        |
| Hardness                      |              | 3.3.2.1               |             | ASTM D2240             |
| Tensile strength              |              | 3.3.2.1               |             | ASTM D412<br>(Die C)   |
| Elongation                    |              | 3.3.2.1               |             | ASTM D412<br>(Die C)   |
| Peel strength                 |              | 3.3.2.1               |             | 4.5.3.2.7              |
| Dielectric strength <u>2/</u> |              | 3.3.2.2               |             | ASTM D149              |

1/ When authorized by the procuring activity shipment may be made prior to the completion of the corrosion test upon receipt of a letter (see 4.5.3.2.9) certifying that the sealing adhesive-sealant shall meet the corrosion test requirements of 3.3.1.1 a.

2/ Quality conformance inspection, for corrosion and dielectric strength, on lots of silicone adhesive-sealant, which are run in a continuous basis, shall be made at least every four months during manufacturer's test inspection."

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## 5.3: Delete and substitute:

"5.3 Marking. In addition to any special marking required by the contract or order herein, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129 for military levels of protection. A label shall be attached to each exterior container of silicone adhesive-sealant and primer (if required) with additional data as follows:

- (a) Number and title of this specification.
- (b) Adhesive-sealant group and type.
- (c) Dates of manufacture and reinspection.

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- (d) Cure time if other than that specified (see 4.5.2.2).
- (e) WARNING: relative to toxicity (see 3.5).
- (f) WARNING: as to fluid resistance (see 6.1.1)
- (g) Label attached to containers of silicone adhesive-sealant shall contain application instructions for adhesive-sealant and primer (see 3.4).
- (h) Each unit container of primer shall be marked as follows:  
"CAUTION: APPLY ONLY IN A WELL VENTILATED AREA. KEEP AWAY FROM HEAT, SPARKS, AND OPEN FLAME"
- (i) WARNING: FAILURE TO USE RECOMMENDED PRIMER ON SOME METALS MAY RESULT IN ADHESION FAILURE."

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6.9 Supersession data. This specification supersedes the following military specifications:

- a. MIL-A-46146A(1) 13 March 1991
- b. MIL-R-47211A(MI) 21 September 1989 - Type III only

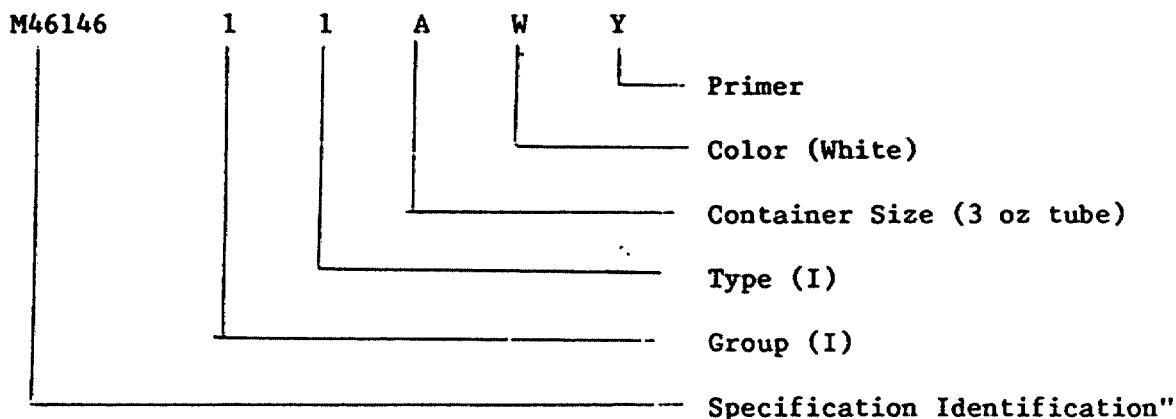
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Table VIII: Delete and substitute:

"TABLE VIII. Military part number coding.

| Group | Code | Type | Code | Container<br>Size | Code | Color       | Code | Primer<br>Required | Code |
|-------|------|------|------|-------------------|------|-------------|------|--------------------|------|
| I     | 1    | I    | 1    | 3 oz tube         | A    | White       | W    | Yes                | Y    |
| II    | 2    | II   | 2    | 5 oz tube         | B    | Translucent | T    | No                 | N    |
| III   | 3    |      |      | 6 oz cartridge    | C    | Gray        | G    |                    |      |
|       |      |      |      | 12 oz cartridge   | D    |             |      |                    |      |
|       |      |      |      | pint              | E    |             |      |                    |      |
|       |      |      |      | 5 gallon pail     | F    |             |      |                    |      |
|       |      |      |      | 50 gallon drum    | G    |             |      |                    |      |

Example:



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"TABLE IX. Comparison data for revision A to revision B

| <u>Revision A</u>                         | <u>Revision B</u>                                                                             |
|-------------------------------------------|-----------------------------------------------------------------------------------------------|
| Type I<br>Type II                         | Group I - General Purpose<br>Type I - Thixotropic paste<br>Type II - Self-leveling liquid     |
| Type III - High Strength<br>Non-corrosive | Group II - High Strength<br>Type I - Thixotropic paste<br><br>Type II - Self-leveling liquid  |
| MIL-R-47211A(MI) Type III                 | Group III - High Temperature<br>Type I - Thixotropic paste<br>Type II - Self-leveling liquid" |

## Custodians:

Army - MR  
Navy - AS  
Air Force - 11

## Preparing activity:

Army - MR

Project 8040-0527

## Review activities:

Army - AR, ME, AT, MI, SM, EA  
Navy - AS, SH, OS  
Air Force - 11, 99

## User activities:

Army - CE, MI, AL  
Navy - OS, SH, YD  
GSA-FSS - 10FTE

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